CHANDNI VERMA

Punjab, India | chandniverma.geo@gmail.com | Portfolio | GitHub | LinkedIn

PROFILE

Remote Sensing and Geospatial Professional with experience in geospatial analysis, satellite image processing, SAR/InSAR, DEM analysis, and Python-based geospatial automation. Skilled in developing reproducible workflows that transform Earth observation data into practical insights.

PROFESSIONAL EXPERIENCE

Junior Research Fellow — 09/2024 – 03/2026
CSIR-Central Scientific Instruments Organisation (CSIO) — Chandigarh, India
A comprehensive Geomorphic Anomaly Investigation over Sutlej Basin, India
- Conducted DEM-based geomorphic and morphotectonic analysis using Python, MATLAB, TopoToolbox, and GIS.
- Developed an automated Python-based DEM profiling workflow for scalable line and swath profile extraction.
- Applied drainage-network, longitudinal profile, and χ-index analysis to investigate landscape evolution and surface processes.
- Conducted field investigations to validate geomorphic interpretations and collect soil samples for OSL dating.
- Prepared geomorphic maps, technical figures, visualizations, and project reports.

Trainee — 10/2024 – 10/2024
Indian Institute of Remote Sensing (IIRS), ISRO — Dehradun, India
Advanced SAR Data Processing for Ground Surface Movement
- Configured Linux-based SAR processing environments and preprocessing workflows for interferometric analysis.
- Performed guided Differential SAR Interferometry (DInSAR) and Small Baseline Subset (SBAS) processing workflows using ASF, SNAP, and ISCE for ground-surface deformation analysis.
- Acquired practical exposure to SAR time-series analysis frameworks, including MintPy, for deformation monitoring and interferometric interpretation.

EDUCATION

M.Tech in Remote Sensing and GIS (Geosciences) — 08/2022 – 07/2024
Indian Institute of Remote Sensing (IIRS), ISRO — Dehradun, India

M.Sc in Geology — 07/2014 – 06/2016
Panjab University — Chandigarh, India

B.Sc in Geology — 07/2011 – 06/2014
Panjab University — Chandigarh, India
Minors: Mathematics and Chemistry

PROJECTS

Land Subsidence Characterization in Northwest India using Sentinel-1 SAR, GRACE, and Groundwater-Level Data
DInSAR | PS-InSAR | Sentinel-1 | GRACE | Aquifer System Compaction
- Analyzed GRACE-derived groundwater-storage anomalies to investigate regional groundwater-depletion trends.
- Processed selected Sentinel-1 SLC image pairs using DInSAR workflows in ENVI/SARscape and performed PS-InSAR time-series analysis of 151 Sentinel-1 SLC scenes (2018–2022) in SARPROZ to derive LOS displacement time series.
- Modeled aquifer-system compaction using piezometric groundwater-level data and compared the modeled deformation with PS-InSAR-derived displacement trends.
- Conducted field investigations to validate InSAR-derived deformation patterns through on-site observations and documentation of deformation-related structural damage.

GeoProfiler / GeoProfilerX: Automated Raster Profile Extraction and Geospatial Analysis
Python | Streamlit | Rasterio | GeoPandas | NumPy | SciPy | Matplotlib
- Developed **GeoProfiler**, an open-source Python tool for automated line and swath profile extraction from DEMs, with terrain visualization and quantitative profile analysis.
- Extended to **GeoProfilerX** to support profiling of diverse single-band geospatial rasters, including DEMs, gravity/magnetic anomalies, and InSAR displacement data.
- Implemented automated raster sampling, profile generation, smoothing, visualization, and data export for quantitative geospatial analysis.
  Web App: geoprofiler.streamlit.app
  GitHub: https://github.com/chandnivermageo/GeoProfilerX

SAR-Based Reservoir Sedimentation and Storage Capacity Assessment using Sentinel-1 GRD Imagery
Sentinel-1 | Google Earth Engine

- Conducted multi-temporal Sentinel-1 SAR analysis for reservoir water-spread and sedimentation assessment using GEE and ArcGIS.
- Integrated SAR-derived water-spread extent with India WRIS reservoir-level data to estimate revised storage capacity and sedimentation impacts.

Snow-Cover Delineation and Change Detection using Landsat 8 and DEM Integration
Python | Landsat 8 | DEM

- Developed an automated Python-based workflow for snow-cover delineation and change detection using a multi-criteria classification approach integrating spectral indices, radiometric thresholds, elevation filtering, and morphological cleanup for improved snow extraction accuracy.
- Performed raster-based spatial analysis, binary change mapping, and elevation-stratified snow distribution assessment using Rasterio, NumPy, SciPy, and Matplotlib.
  GitHub: github.com/chandnivermageo/Snow-Cover-Change-Detection

SOFTWARE & TOOLS

ArcGIS • QGIS • SNAP • ENVI/SARscape • SARPROZ • Google Earth Engine • ERDAS IMAGINE • Google Earth Pro

TECHNICAL SKILLS

Remote Sensing & Earth Observation: Geospatial analysis, SAR/InSAR, PS-InSAR time-series analysis, satellite image processing, change detection.

Terrain & Geomorphic Analysis: DEM processing, terrain analysis, watershed and drainage delineation, topographic profile extraction, fluvial geomorphology, terrace mapping, morphotectonic analysis.

Programming & Scientific Computing: Python, MATLAB, R, NumPy, SciPy, Rasterio, GeoPandas, Matplotlib, TopoToolbox, Streamlit, geospatial workflow automation.

WORKSHOP & CONFERENCES

- National Aquifer Mapping (NAQUIM) and Groundwater Data Dissemination for Sustainable Management, CGWB. Workshop, March 2026.
- Observing Ground Deformation from Space and in the Field, CSIR-CSIO. International Workshop, December 2024.
- IIRS Academia Meet (IAM-2024). Poster presentation, *Land Subsidence Characterization in Northwest India using GRACE and SAR Data*, 2024.
- 6th World Congress on Disaster Management (2023). Abstract selected, *Reservoir Sedimentation Assessment of Tehri Reservoir Using SAR Data.*
- 10th Chandigarh Science Congress (CHASCON-2016). Poster presentation, 2016.

LANGUAGES

English, Hindi, and Punjabi

ACHIEVEMENTS AND SCHOLARSHIP

- GATE 2025 (GE-Geomatics): All India Rank 153
- GATE 2024 (GE-Geomatics): All India Rank 239
- Recipient AICTE M.Tech GATE Scholarship (2022-2024)
- GATE 2022 (GG-Geology & Geophysics): All India Rank 763

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- Winner (Mixed Table Tennis & Badminton), CSIR–CSIO Sports Tournament, 2025.
- Key team member in organizing the International Conference on Land Surface Deformation at CSIR-CSIO (Dec 2024).
- Held leadership roles as General Secretary (2015-16) and Cultural Secretary (2013-14) at Pick & Hammer Club, Panjab University.

INTERESTS

Singing, Guitar, Badminton and Table Tennis